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1. Introduction & Background Study

Key Background Concepts

- **Data Compression:** Minimizes digital storage footprints and network transmission bandwidth across systems.
- **Lossless Guarantee:** Ensures 100% exact, bit-for-bit file reconstruction without any distortion or loss of data.
- **Shannon & Huffman Foundation:** Reaches Shannon entropy lower bound $H(X)$ using optimal, greedy prefix binary codes.

2. Problem Statement & Objectives

Problem Statement

- Raw uncompressed text and log files incur heavy storage and bandwidth costs.
- Existing industrial tools operate as opaque, closed-source “black boxes”.
- Lack of lightweight tools offering verified byte-for-byte fidelity.

Key Project Objectives

- Build a pure, lossless Huffman compression and decompression engine.
- Implement min-heap priority queue for optimal prefix tree synthesis.
- Guarantee 100% integrity via automated cryptographic SHA-256 checks.

3. Literature Review & Benchmark

Theoretical Foundation & Benchmarks

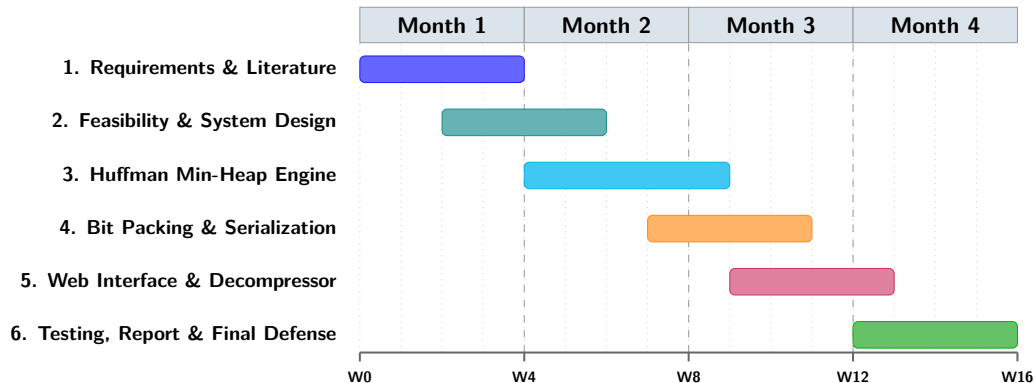
- **C. E. Shannon (1948)**: Proved the mathematical limits of data compaction based on source entropy $H(X)$.
- **D. A. Huffman (1952)**: Devised bottom-up greedy binary tree synthesis for minimum-redundancy prefix codes.
- **System Advantage**: Provides transparent, canonical prefix encoding without complex, proprietary black-box layers.

4. Methodology & Huffman Algorithm Workflow

Three-Phase Compression & Decompression Pipeline

- **Frequency Profiling & Min-Heap:** Count all 256-byte frequencies and assemble a min-heap priority queue to build the binary tree.
- **Codeword Assignment & Bit Packing:** Traverse branches (left='0', right='1') to generate prefix codes and pack bits into 8-bit bytes with header metadata.
- **Restoration & Verification:** Parse the header tree, decode the bitstream back to original bytes, and verify SHA-256 hash match.

5. Project Timeline (4-Month Gantt Chart)



- **4-Month Schedule:** Month 1 (Planning) → Month 2–3 (Implementation) → Month 4 (Testing & Defense).

6. Expected Outcomes & Conclusion

Project Deliverables

- **Lossless Software Tool:** Pure canonical Huffman coding engine with self-contained archive packaging.
- **Proven Space Reduction:** 20% to 60% storage savings on text, CSV, logs, and application code files.
- **Verified Data Integrity:** Guaranteed 100% byte-for-byte lossless reconstruction confirmed by SHA-256.

Thank You!

Questions & Answers

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